# HospitAI

A hospital data analytics platform that combines
patient grouping, clinical sequence mining,
and AI-generated interpretation

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# Outline


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**Overview**

In short, **HospitAI** is an **end-to-end** analytics pipeline that ingests raw EHR data, applies **unsupervised patient clustering** and **clinical sequence mining**, and provides **AI-generated** interpretations to support data-driven hospital insights.

# 1

# Introduction and context

# Context and challenges (EHR data)

- EHR (Electronic Health Record) systems gather huge volumes of clinical data, but most of it stays **unused**. Also, querying it requires SQL expertise that most medical staff lack.
- A global analysis on that data hides **patient diversity**: an acute case is very different from the many conditions of an elderly patient.
- Medical care is strongly **time-based**. The order of actions matters more than how often they happen.
- Classic algorithms return lists of abstract codes that clinicians cannot interpret quickly.

**HospitAI** addresses these challenges with: patient grouping (clustering), clinical sequence mining, an AI interpretation layer, and an AI Chat Agent.

# Main objective and sub-objectives

## Main objective

Build the **end-to-end** **HospitAI** pipeline that turns raw clinical data into explainable medical decisions.

## Sub-objectives

1. **Patient grouping:** cluster patients into homogeneous groups using K-Prototypes on mixed-type data.
2. **Sequence mining:** extract and rank treatment patterns over time using PrefixSpan and GML score.
3. **AI verbalization:** generate readable clinical summaries per cluster via an LLM verbalizer.
4. **SQL agent:** allow patients or medical staff to query clinical data in plain English via a LangGraph agent; also responds to general health-related questions.

# 2

# Related work

# Related work: what others did

| Category | Papers | What they do |
|---|---|---|
| Surveys | Rosa (2022), Flothow (2023) | Rosa distinguishes sequence mining from process mining in pathway analysis. Flothow separates full-trajectory clustering from subsequence pattern mining. |
| Clustering and mining | Najjar (2018), Baudrier (2022), Vogt (2017) | Najjar clusters individual visits with K-Prototypes then derives pathways. Baudrier links trajectory clustering to outcomes in kidney cancer. Vogt clusters heart-failure pathways from insurance claims. |
| Sequence mining | Perer (2015) | Mines frequent event sequences from EMRs and links them to patient outcomes via visual analytics. |
| Patient clustering | Yang (2024), Aljohani (2024) | Yang clusters EHR patients to reveal health risks. Aljohani evaluates eight clustering methods for patient stratification. |
| Process mining | Murazzano (2026) | Applies process mining to chronic respiratory patients, observing high variability in care processes. |

# Related work: gaps and our contribution

| Category | Gap |
|---|---|
| Surveys | Overview only; no unified implementation |
| Clustering and mining | Visit-level grouping; single-disease scope; steps applied in a different order |
| Sequence mining | Frequency only; no clinical ranking metric to filter trivial patterns |
| Patient clustering | No sequential analysis component |
| Process mining | No AI interpretation or verbalization layer |

## What HospitAI adds

- **Patient-level** mixed clustering with K-Prototypes across **22,284** patients, not visit-level
- Cross-domain applicability, not limited to a single disease or patient type
- **Clinically ranked** sequences via the GML score, which rewards genuine statistical association beyond raw frequency
- **LLM verbalization** that generates readable cluster summaries for both clinicians and patients
- **Natural-language SQL agent** built on LangGraph, enabling clinical queries without SQL expertise

# 3

# The HospitAI analytics pipeline

# Pipeline architecture and ETL (Stage 1)

- The architecture is split in two: an offline analysis pipeline in **Python** that processes raw EHR data and produces clustering and sequence mining results, and a fast web application (**Next.js and FastAPI**) that displays those results and allows medical staff to perform CRUD operations on the data.
- **Stage 1 - ETL (extract, transform and load):** Synthea complex synthetic EHR dataset (**22,284** patients), used to ensure full **privacy compliance** (GDPR). Missing values filled by **mode** / **median**.

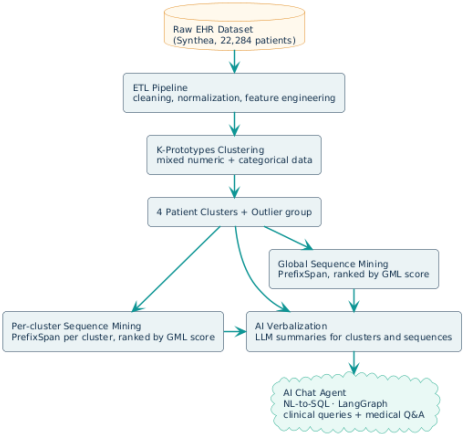



Fig. 1. The HospitAI pipeline

# Patient Grouping (Stage 2)

- **Why K-Prototypes:** it handles mixed data (numeric and categorical) natively. K-Means fails on categorical attributes; K-Modes fails on numeric ones.
- Many features were **derived** from raw EHR data rather than read directly. For example, the **clinical severity score** is computed from condition count, medication load, and procedure frequency.
- **K=4** was selected via an **elbow plot** on the Gower inertia curve, which showed a clear inflection point at 4 clusters.
- At cluster creation, the **AI layer** receives the feature values via prompt and generates a **title**, **insights**, and **recommendations** tailored to each patient group.

**Huang mixed distance ($\gamma$ = categorical weight)**

$$d(X,Y) = \sum_{j=1}^{p}(x_j - y_j)^2 + \gamma \sum_{j=p+1}^{m} \delta(x_j, y_j)$$

**Outlier detection**

$$z = \frac{x - \mu}{\sigma} > 6 \;\Rightarrow\; \text{class } (-1)$$

HospitAI - Patient Clusteri

# Sequence Mining (Stage 3)

- Patient timelines are built over a **2-year window**; consecutive duplicates removed to avoid noise.
- **PrefixSpan** extracts frequent subsequences with minimum support **1%**.
- **Why not frequency alone:** common events carry no clinical signal. The **GML score** measures genuine statistical association between events.
- A **diversity bonus** rewards patterns mixing medications, procedures, and observations.
- Mining runs at **two levels**: globally and per cluster.

**GML score**

$$\mathrm{GML}(p) = \frac{s(p)/n}{\left(\prod_{i \in p} s(i)/n\right)^{1/k}}$$

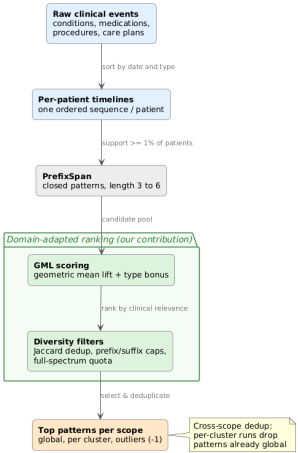



Fig. 2. The sequence mining pipeline

HospitAI - Patient Cluste

# The AI layer and the SQL agent (Stage 4)

## AI Verbalization

- Clusters are interpreted **automatically at creation**: the LLM generates a **title**, **description**, **insights**, and **recommendations** from precomputed feature values.
- Interpretation can also be **triggered on demand**.
- Two output modes: **medical-oriented** (clinicians) and **patient-oriented** (plain language).
- **Sequences** can also be verbalized by the AI layer.

## SQL Agent

- Translates natural language to SQL; behavior adapts to **user role**.
- Answers general medical questions with a disclaimer: responses are **not a substitute for a doctor**.
- Out-of-scope questions are **refused**.

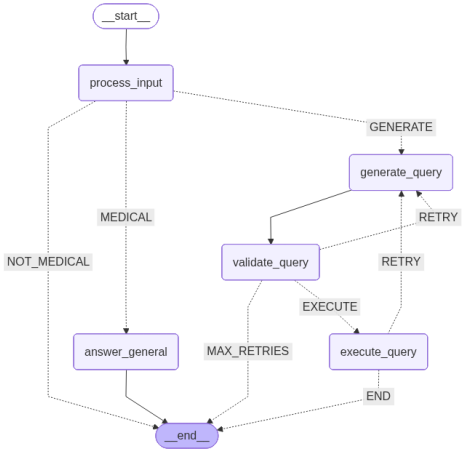



Fig. 3. The agent state machine (LangGraph)

# 4

# Simulations and experimental results

# Simulation: experimental setup and data

- After ETL, in Stage 2, most features that will be used for clustering are extracted through **SQL aggregation queries** over the raw EHR tables.
- Some are **derived** from others via feature engineering, e.g., tabulating clinical events into scores or ratios. Below there are listed all the features.

| Group | Features |
|---|---|
| Demographics | age, gender, race, ethnicity, marital status |
| Encounter utilization | inpatient, outpatient, emergency, wellness, ambulatory, urgentcare, SNF, hospice counts; dominant encounter class; total encounter duration (hours); days since last encounter; routine care count, acute care count, severe care count *(derived)*; preventive-to-reactive ratio *(derived)*; out-of-hours flyer; has inpatient history; readmission risk tier |
| Conditions | total conditions, active conditions count, condition burden *(derived: low/medium/high)*; has_diabetes, has_hypertension, has_heart_disease, has_cancer, has_kidney_disease, has_mental_health, respiratory disease profile |
| Medications and procedures | total medications, lifetime medication days, is_polypharmacy, medication intensity *(derived: low/medium/high)*, pain management tier; total procedures; clinical severity score *(PCA-derived from procedures, medications, and active conditions)* |
| Vitals and observations | avg BMI, BMI type *(derived: Underweight to Obese III)*; avg systolic BP, BP risk category *(derived)*; avg glucose, glucose risk category *(derived: Normal/Prediabetic/Diabetic)*; QALY, DALY |
| Behavioral and risk | substance use profile, smoking status, cholesterol risk category, pregnancy history; distinct providers seen; dominant care plan |

# Results: the obtained clusters

- **Cluster 0 (Young Injury):** young patients, acute trauma, emergency and urgentcare encounters, low chronic burden.
- **Cluster 1 (Maternity):** predominantly female, maternal care plans, low medication load, wellness encounters.
- **Cluster 2 (Older High Risk):** elderly, polypharmacy, hypertension, diabetes, heart disease, long encounter duration.
- **Cluster 3 (Pediatrics):** children, minimal active conditions, low procedure counts, wellness dominant.
- **Cluster ($-1$) (Outliers):** extreme cases (oncology, dialysis, terminal care), stored separately for independent analysis.

**Why the clusters are good**

- Each cluster maps to a **recognizable patient type** that a clinician can immediately interpret.
- The separation is driven by **clinical behavior**: conditions, medications, and encounter patterns, not superficial demographics.
- Each group has a **distinct care trajectory** that justifies different treatment and prevention strategies.

# Results: cluster profiles

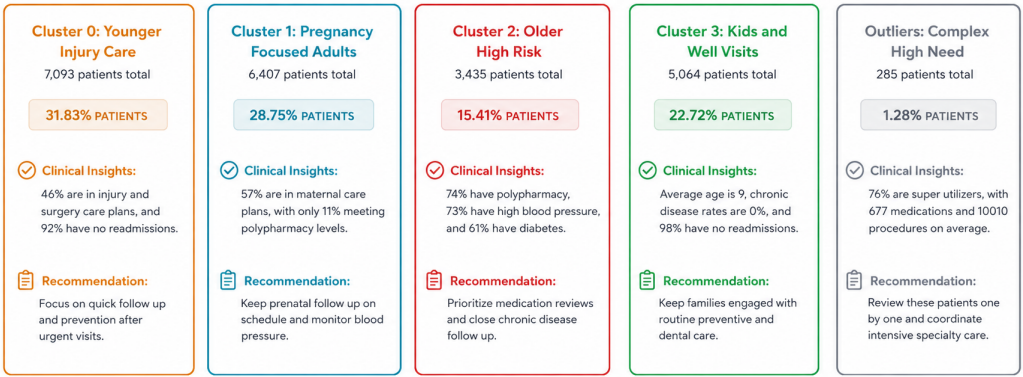



Fig. 4. The obtained cluster profiles - fully AI interpreted

# Results: meaningful clinical sequences (length 4–5)

PrefixSpan (min. support 1%), sequences of length 4–5 ranked by **GML score**. Top non-dental patterns per cluster:

| | | |
|---|---|---|
| **C0 (Young Injury)** | Laceration → Wound care → Suture open wound → Acetaminophen | 0.30 |
| **C0 (Young Injury)** | Acute bronchitis → Respiratory therapy → Resp. measurement → Acetaminophen | 0.44 |
| **C1 (Maternity)** | Tubal ligation (history) → Minor surgery care → Bilateral tubal ligation → Postoperative care | 0.81 |
| **C1 (Maternity)** | Miscarriage → Normal pregnancy → Standard pregnancy test → Fetal ultrasound | 0.37 |
| **C2 (Older High Risk)** | Fluticasone/Salmeterol inhaler → Albuterol → Pulmonary rehabilitation → Spirometry | 0.70 |
| **C2 (Older High Risk)** | Oxygen administration → Prone positioning → Furosemide inj. → Heart failure education | 0.61 |
| **C3 (Pediatrics)** | Fracture of bone → Fracture of clavicle → Orthopedic admission → Clavicle X-ray | 0.62 |
| **C3 (Pediatrics)** | Fracture of ankle → Fracture of bone → Fracture care → Bone immobilization → Ankle X-ray | 0.44 |
| **C(−1) (Outliers)** | Renal dialysis → Epoetin Alfa → Renal dialysis → Epoetin Alfa | 0.77 |
| **C(−1) (Outliers)** | Chemo+radiation → Cisplatin → Paclitaxel → Chemo+radiation → Hospice care | 0.72 |

Table 1: Suggestive sequences of length 4–5 per cluster (GML score)

# Results: cluster interpretation and Chat Agent

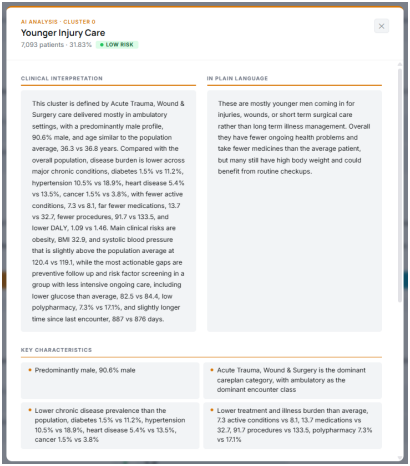



Fig. 5. AI-generated cluster interpretation

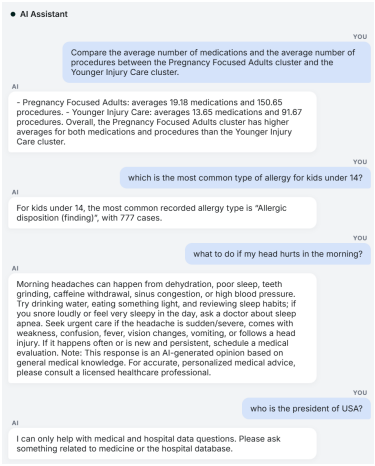



Fig. 6. Chat Agent - example interaction

# 5
# Discussion and performance review

# Discussion: benchmark logic and results validation

- **629** total runs across **60** feature configurations (v1–v60). Each config defines a distinct set of numerical and categorical features.
- Per config, a **nested loop** varied: $k \in \{3, 4, 5, 6\}$ and Z-score threshold $\in \{4, 6\}$. For the best candidates, gamma $\in \{\text{auto}, 0.07, 0.13, 0.17\}$ and random seed $\in \{7, 42, 100\}$ were also tested.
- Each run was evaluated with three metrics: **Gower Silhouette**, **Calinski–Harabasz**, and **Davies–Bouldin**, combined via a min-max normalized empirical score. Runs that split patients only by gender were excluded.
- Runs were sorted by score; the best configuration per $k \in \{4, 5, 6\}$ was selected (see table).
- For **sequence mining**: patterns ranked by **GML** (geometric mean lift). A **diversity bonus** $(1 + 0.3\,(t - 1))$ rewards patterns spanning multiple event types.

**Optimization score (min-max normalized, empiric)**

$$\text{Score} = 0.5\,\text{Gower} + 0.3\,CH_n + 0.2\,DB_{inv}$$

| Config | $k$ | Gower | Score |
|---|---|---|---|
| **v55** | **4** | **0.350** | **0.82** |
| v40 | 5 | 0.324 | 0.74 |
| v60 | 6 | 0.287 | 0.70 |

**v55 feature set ($k = 4$)**

| | |
|---|---|
| **Numerical (5)** | age, active conditions, total medications, total procedure prev./reactive ratio |
| **Categorical (8)** | gender, pregnancy history, h disease, diabetes, hyperten condition burden, BMI type, dominant care plan |

Table 2: v55 feature set. Results above are based on these features.

# 6

# Conclusions and future work

# Conclusions and future work

**Contribution**

HospitAI shows that **joining** mixed clustering with ranked mining turns the noise in EHR data into **clear medical paths**.

**What the results enable**

- **Clusters** support risk stratification and targeted resource allocation per patient group.
- **Sequences** can seed or validate clinical treatment protocols and guidelines.

**Limitations**

- **Synthetic** data (Synthea).
- It relies on an external language model, which raises hospital governance concerns.

**Future work**

- Direct clinical review with specialist doctors.
- Testing on **real**, anonymized databases.

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# Q & A

Thank you for your attention!

# 7

# Appendix: formulas

# Appendix I: clustering and ETL pipeline

## K-Prototypes distance (Huang)

$$d(X,Y) = \sum_{j=1}^{p}(x_j - y_j)^2 + \gamma \sum_{j=p+1}^{m} \delta(x_j, y_j), \qquad \delta(x_j, y_j) = \begin{cases} 0, & x_j = y_j \\ 1, & x_j \neq y_j \end{cases}$$

## Outlier detection (Z-score, post-log)

$$z = \frac{x - \mu}{\sigma} \;\Rightarrow\; z > 6 \Rightarrow \text{class } (-1)$$

## Scaling and normalization

$$X_{norm} = \frac{\ln(X) - \min(\ln X)}{\max(\ln X) - \min(\ln X)}$$

# Appendix II: sequence mining and evaluation

**Geometric Mean Lift (GML), base form**

$$\mathrm{GML}(p) = \frac{s(p)/n}{\left(\prod_{i \in p} s(i)/n\right)^{1/k}}$$

**Final GML (with a diversity bonus, $t$ = number of event types)**

$$\mathrm{GML}_{final}(p) = \mathrm{GML}(p) \cdot \left(1 + 0.3\,(t-1)\right)$$

**Weighted linear optimization (hyperparameter tuning)**

$$\mathrm{Score}_{config} = 0.5 \cdot \mathrm{Gower}_{norm} + 0.3 \cdot CH_{norm} + 0.2 \cdot DB_{inv_norm}$$